TEXORA · LECTURE SHEET

TQM Principles in Practice

The principles that run a TQM organisation day to day — quality statements, the customer chain from focus to retention, continuous process improvement through PDCA, 5S and Kaizen, and partnership with suppliers including supplier selection and rating.

TE-4108 · QUALITY ASSURANCE AND MANAGEMENT IN TEXTILES · TOPIC 2 · LEVEL 4 TERM 1

1. Quality statements: vision, mission and policy

TQM begins with declared intent at three levels. A **vision statement** describes the future the organisation aims at ("to be the most trusted fabric source in our market"). A **mission statement** states its present purpose and scope. A **quality policy** is the management's formal commitment — what the organisation will do about quality (meet customer requirements, comply with standards, improve continually) — and under ISO 9001 it must be documented, communicated and used to set measurable **quality objectives** in each department. Statements earn their wall space only when objectives, reviews and resources trace back to them; otherwise they are decoration, and the workforce can tell the difference within a week.

2. The customer chain: focus, orientation, satisfaction, complaints, retention

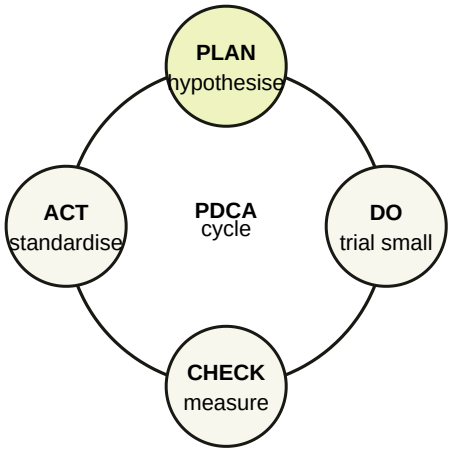
Customer focus and **customer orientation** mean organising decisions around the customer's requirements — including the **internal customer**: weaving is spinning's customer, dyeing is weaving's, the sewing line is the cutting room's. Each handover has requirements, and each supplier in the chain owes the next process a conforming input. **Customer satisfaction** is the measured gap between expectation and experience: a buyer satisfied with today's shade tolerance will expect it tightened next season, so satisfaction is a moving target that must be re-measured (surveys, claim rates, repeat-order share, audit scores).

Customer complaints are data with the postage paid by the customer: each is logged, analysed for root cause (Topic 3 tools), answered within a committed time, and closed only when the corrective action is verified — a complaint system that apologises without analysing trains customers to leave quietly. **Customer retention** follows: keeping a buyer is far cheaper than winning one, and retention metrics (repeat orders, share of the buyer's volume, relationship length) are therefore quality metrics. In export

textiles, retention is the business: a retained buyer funds capacity planning; a churning one makes every season a cold start.

3. Continuous process improvement and the PDCA cycle

Continuous process improvement holds that every process can be made better in small, permanent steps, forever — the opposite of improvement-by-occasional-campaign. Its engine is the **PDCA (Deming) cycle**:



Plan → Do → Check → Act, clockwise, then around again: each turn raises the standard and locks it.

Fig. 1 — The PDCA cycle. "Act" both standardises a success and starts the next loop — improvement is a wheel, not a staircase with a top.

- **Plan** — select a problem from data, analyse causes, propose a countermeasure and predict its effect.
- **Do** — trial the countermeasure small-scale (one line, one machine, one week).
- **Check** — measure against the prediction with the same data definitions as the baseline.
- **Act** — if it worked, standardise it (documents, training, control plan); if not, the learning still feeds the next Plan. Then the cycle repeats on the next problem.

4. 5S — workplace organisation as a quality foundation

5S organises the workplace so that abnormality is visible at a glance — the precondition of process control:

S	MEANING	TEXTILE FLOOR EXAMPLE
Seiri — Sort	Remove what is not needed	Clear obsolete patterns, broken accessories and dead stock from the sewing floor
Seiton — Set in order	A place for everything, labelled	Shadow boards for tools; marked locations for cones by count and shade

Seiso — Shine	Clean — and inspect while cleaning	Machine cleaning schedules that double as leak and wear inspection
Seiketsu — Standardise	Make the first three S the routine	Posted zone standards with photographs of the correct state
Shitsuke — Sustain	Discipline and audit to hold the gains	Monthly 5S audits with scores by section

5. Kaizen

Kaizen ("change for the better") is the philosophy behind continuous improvement: everyone, every day, improving their own process in small steps, with management's role being to invite, evaluate and implement suggestions quickly. Kaizen events (focused improvement workshops on one area over a few days) complement daily kaizen. Its textile power is that the people who see a defect's birthplace daily — operators, mechanics, helpers — hold most of the usable knowledge; a suggestion system that answers every suggestion within days, implements the workable ones visibly, and feeds results back, converts that knowledge into improvement. A suggestion box whose contents vanish is worse than none.

6. Supplier partnership, selection and rating

TQM extends the customer chain backwards: the mill is its suppliers' customer, and it should behave like the customer it wishes for — clear specifications, stable forecasts, prompt feedback, and **partnering**: long-term relationships in which supplier and mill improve together (joint shade-development with a dye supplier, bale-quality feedback to a ginner) instead of annually re-tendering on price alone. Adversarial buying buys the cheapest input and the most expensive production.

Supplier selection evaluates capability before price: quality system and lab facilities, process control evidence, delivery reliability, technical support, financial stability, and trial-lot performance. **Supplier rating** then scores approved suppliers periodically on weighted criteria:

WORKED EXAMPLE — SUPPLIER RATING OF TWO YARN SUPPLIERS

Weights: quality 40 %, delivery 30 %, price competitiveness 30 %. Scores out of 100.

Supplier A: quality 85, delivery 90, price 70 → 0.40 × 85 + 0.30 × 90 + 0.30 × 70 = 34 + 27 + 21 = **82.0**.

Supplier B: quality 92, delivery 75, price 85 → 0.40 × 92 + 0.30 × 75 + 0.30 × 85 = 36.8 + 22.5 + 25.5 = **84.8**.

Decision: **B rates higher overall** despite weaker delivery — its quality and price strength outweigh A's reliability at these weights. The rating does not end the conversation: B's delivery

> score triggers a development discussion (order scheduling, buffer stock), and the weights themselves are policy — a mill burned by late deliveries would raise the delivery weight and could reverse the ranking. Ratings make such trade-offs explicit and auditable instead of personal.

Glossary

Quality policy — management's documented commitment to quality, basis for quality objectives. **Internal customer** — the next process in the chain, owed a conforming input. **Customer retention** — keeping existing customers, tracked by repeat business. **PDCA** — Plan–Do–Check–Act improvement cycle. **5S** — Sort, Set in order, Shine, Standardise, Sustain. **Kaizen** — continuous small-step improvement by everyone. **Partnering** — long-term collaborative supplier relationships. **Supplier rating** — periodic weighted scoring of approved suppliers.

Key points

• Vision, mission and quality policy matter only when objectives and resources trace to them.
• Every process has a customer; complaints are root-cause data, and retention is a quality metric.
• PDCA turns improvement into a loop: trial small, check honestly, standardise what works.
• 5S makes abnormality visible; Kaizen makes improvement everybody's daily work.
• Supplier rating worked: B 84.8 beats A 82.0 at 40/30/30 weights — and the weights are a stated policy choice.

Practice

1. With the same weights as the worked example, rate Supplier C: quality 78, delivery 95, price 88. Rank A, B and C.
2. Write one PDCA loop for the problem "weft breaks rising on one loom group" — one line per step, with the Check measure specified.
3. Explain how treating the dye house as the weaving department's "supplier" changes what weaving may legitimately demand — and what dyeing may demand in return.
4. A factory's complaint log shows fast, courteous replies but the same shade complaint recurs monthly. Which principle is missing and what changes?

Answer key

1. C = 0.40 × 78 + 0.30 × 95 + 0.30 × 88 = 31.2 + 28.5 + 26.4 = 86.1. Ranking: C 86.1 > B 84.8 > A 82.0 — C's excellent delivery and good price outweigh the lowest quality score at these weights; a

quality-weighted policy review would be the follow-up question, since C's quality (78) trails both rivals.

2. Plan: from loom-stop records (Topic 5, TE-4107) the group shows weft breaks 2× the shed average; likely causes — weft package quality, pirn/bobbin condition, weft tension settings; countermeasure: quarantine the suspect weft lot and reset tension on two looms. Do: run the two reset looms on the suspect lot and two control looms as-is for one week. Check: weft-break stops per loom-hour on trial vs control looms, same fabric. Act: if trial looms improve materially, standardise the tension setting and lot-quarantine rule in the weaving control plan; if not, return to Plan with the tension hypothesis rejected and test package quality next.
3. As a supplier, dyeing owes weaving (its customer) fabric that conforms to agreed requirements — shade, GSM, width, delivery date — and honest early warning when it cannot. In return, a customer owes its supplier what good customers owe: clear, stable specifications, realistic lead times, grey fabric that itself conforms (weaving's output quality is dyeing's input quality), and feedback data rather than blame. The internal customer chain is reciprocal: every department is simultaneously a customer entitled to conforming input and a supplier obliged to provide it.
4. Root-cause corrective action (the Check–Act discipline of PDCA applied to complaints). Courteous replies are appraisal-stage behaviour — they manage the symptom. The missing step is analysing recurrence: log complaints by cause, take the recurring shade cause through cause analysis, change the process (e.g. lab-to-bulk correlation, dosing control), verify over following months that the cause's frequency falls, and close complaints only on that evidence.

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